IT3010 - NDM

Practical 02- Lab Report

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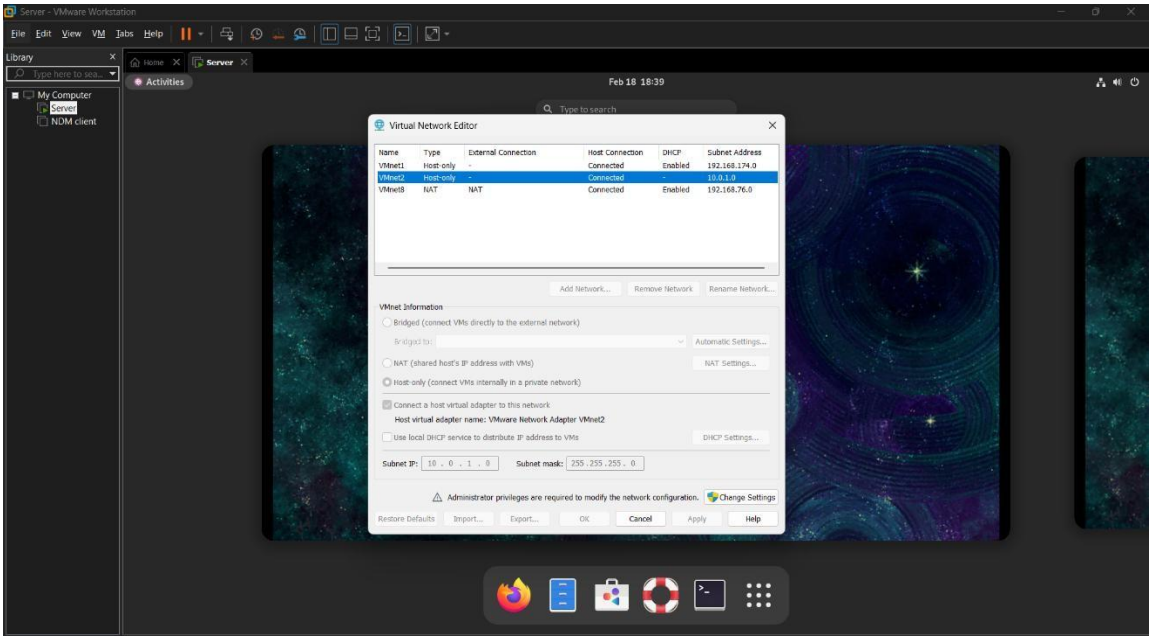
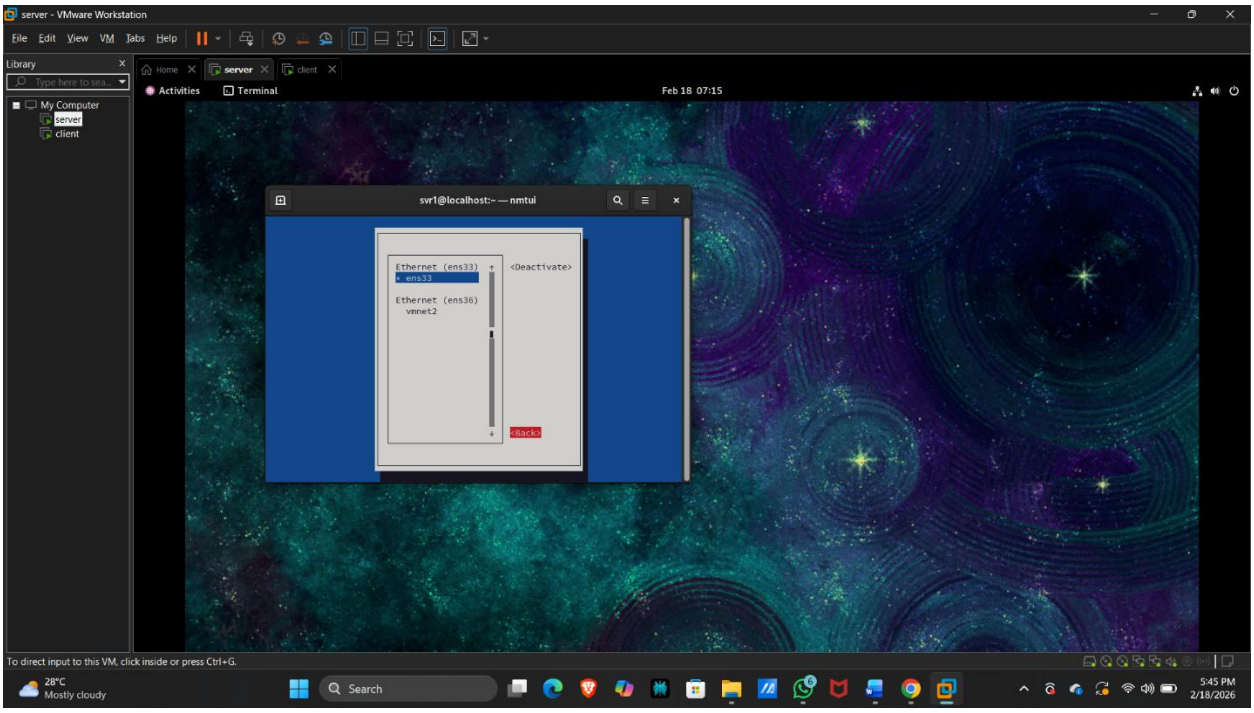
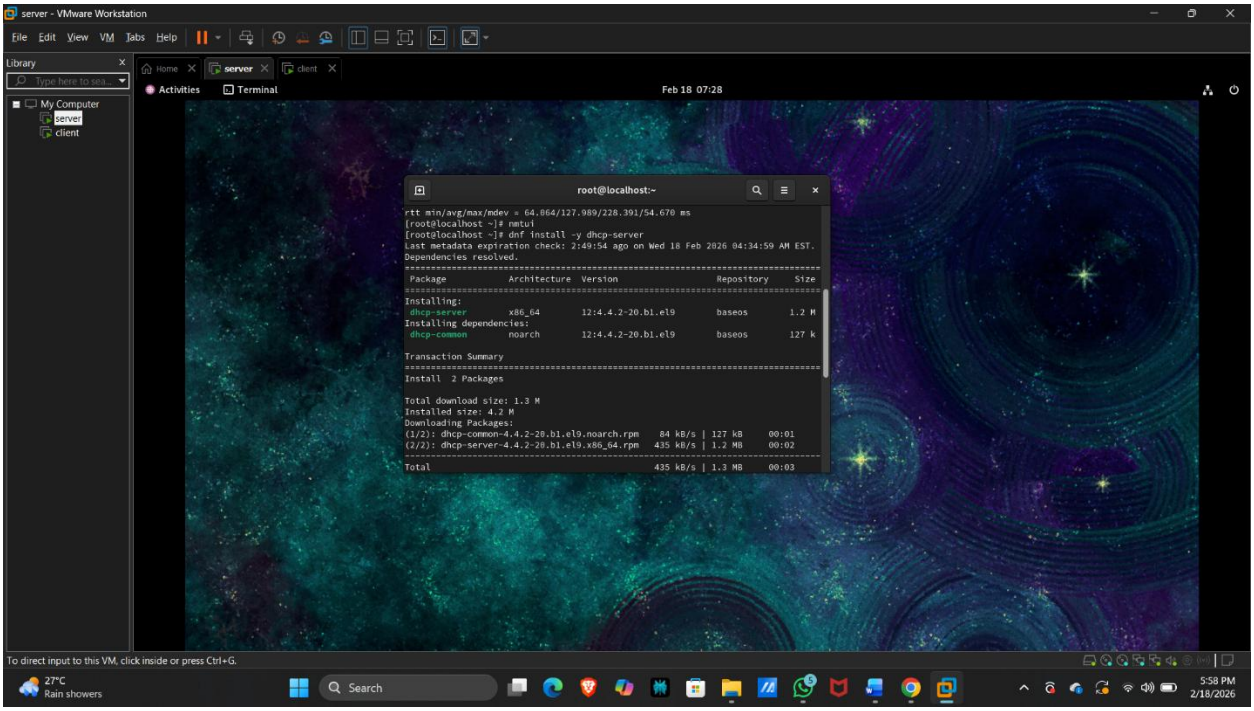
DHCP installation and configuration

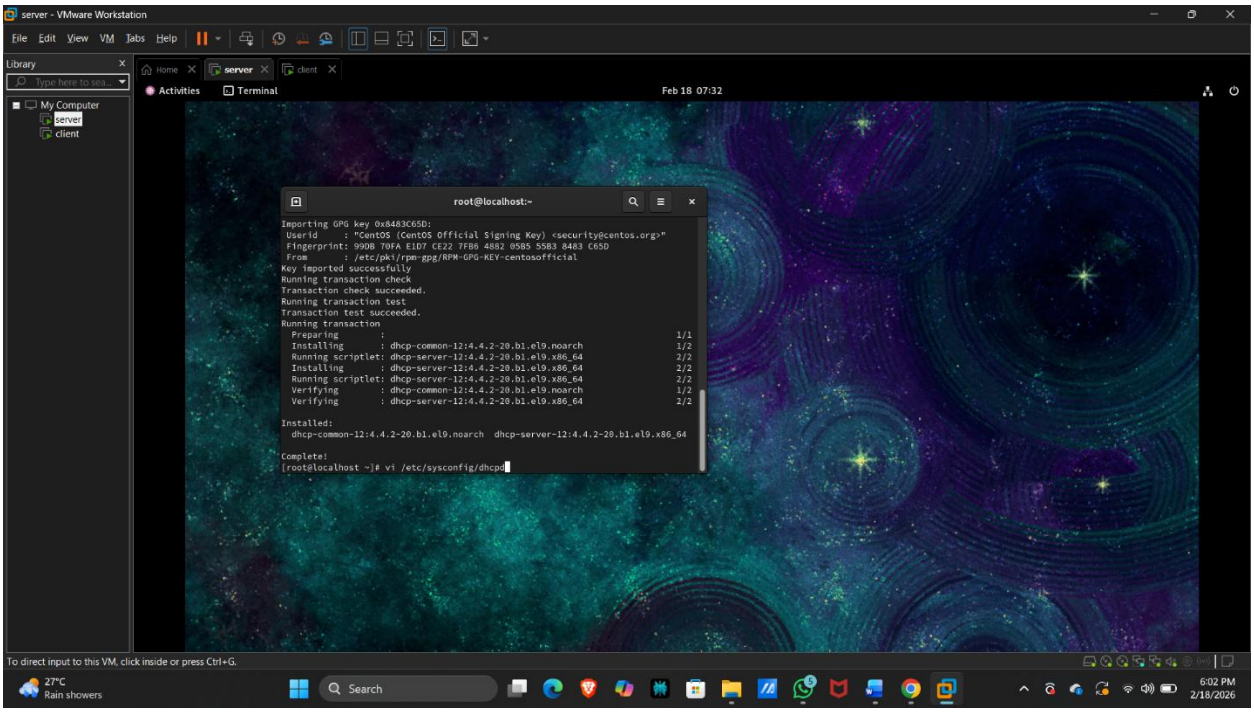
1.Disable VMware DHCP on VMnet2

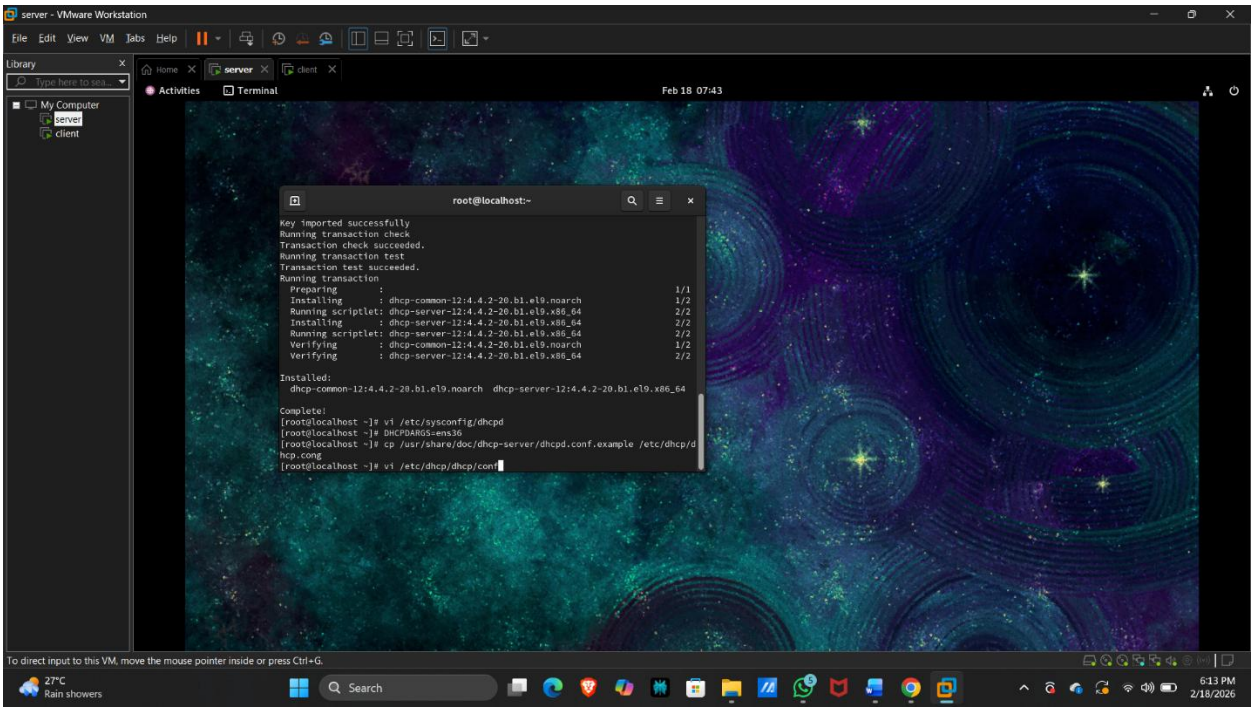


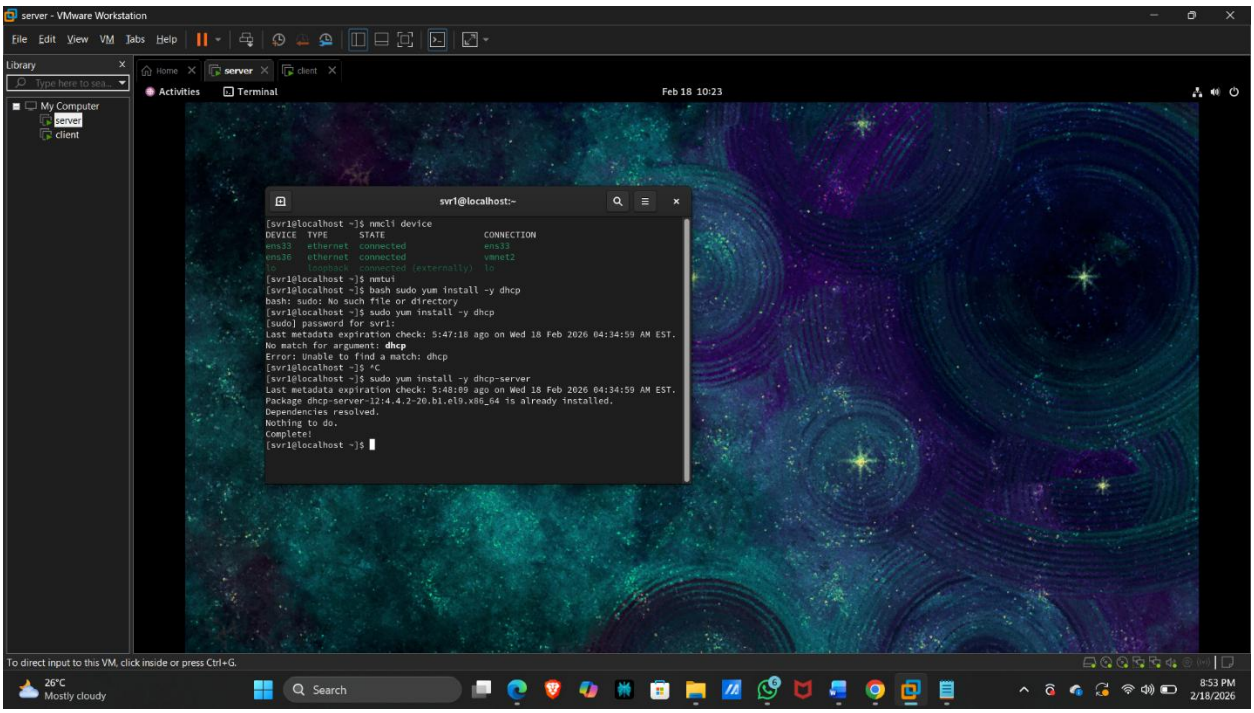


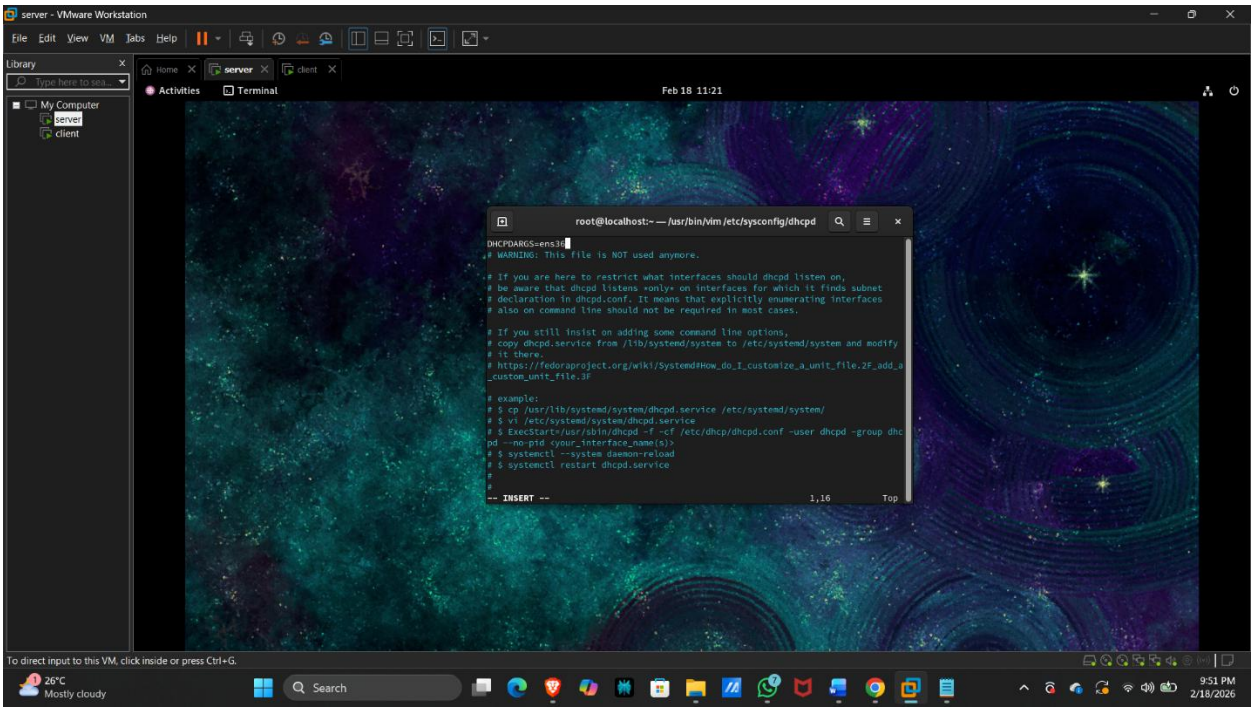
2. Install DHCP Server on CentOS Stream 9



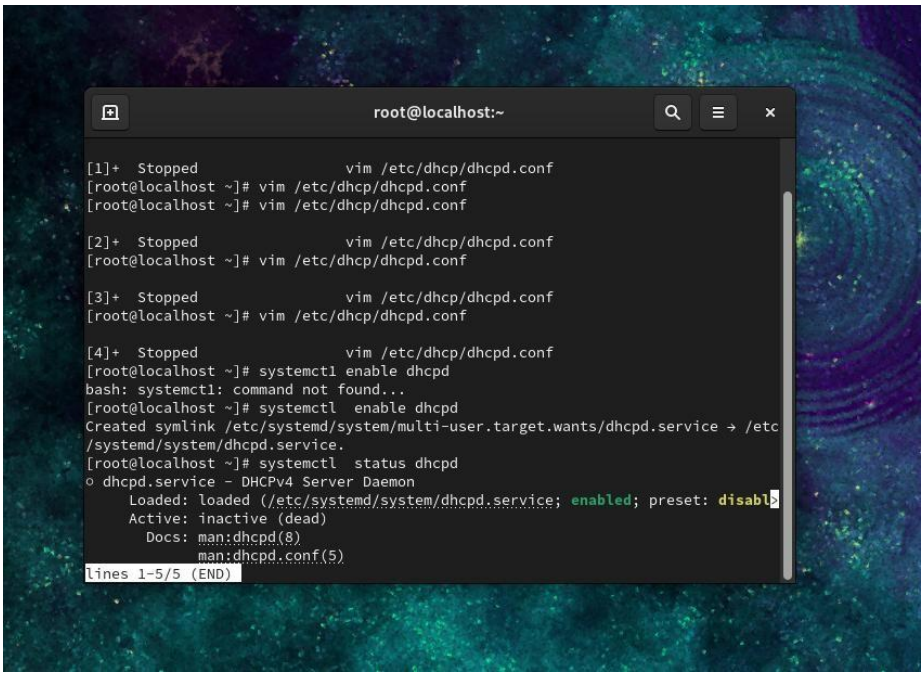



server - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
server
client
Home
server
client
Activities
Terminal
Feb 18 07:43
root@localhost:~
Key imported successfully
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : 1/1
Installing : dhcp-common-12:4.4.2-20.b1.el9.noarch 1/2
Running scriptlet: dhcp-server-12:4.4.2-20.b1.el9.x86_64 2/2
Installing : dhcp-server-12:4.4.2-20.b1.el9.x86_64 2/2
Running scriptlet: dhcp-server-12:4.4.2-20.b1.el9.x86_64 2/2
Verifying : dhcp-common-12:4.4.2-20.b1.el9.noarch 1/2
Verifying : dhcp-server-12:4.4.2-20.b1.el9.x86_64 2/2
Installed:
dhcp-common-12:4.4.2-20.b1.el9.noarch dhcp-server-12:4.4.2-20.b1.el9.x86_64
Complete!
[root@localhost ~]# vi /etc/sysconfig/dhcpd
[root@localhost ~]# DHCPDARGS=ens36
[root@localhost ~]# cp /usr/share/doc/dhcp-server/dhcpd.conf.example /etc/dhcp/dhcp.conf
[root@localhost ~]# vi /etc/dhcp/dhcp/conf
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
27°C
Rain showers
Search
6:13 PM
2/18/2026


server - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
server
client
Home
server
client
Activities
Terminal
Feb 18 10:23
svr1@localhost:~
[svr1@localhost ~]$ nmcli device
DEVICE TYPE STATE CONNECTION
ens33 ethernet connected ens33
ens36 ethernet connected vmnet2
lo loopback connected (externally) lo
[svr1@localhost ~]$ nmtui
[svr1@localhost ~]$ bash sudo yum install -y dhcp
bash: sudo: No such file or directory
[svr1@localhost ~]$ sudo yum install -y dhcp
[sudo] password for svr1:
Last metadata expiration check: 5:47:18 ago on Wed 18 Feb 2026 04:34:59 AM EST.
No match for argument: dhcp
Error: Unable to find a match: dhcp
[svr1@localhost ~]$ ^C
[svr1@localhost ~]$ sudo yum install -y dhcp-server
Last metadata expiration check: 5:48:09 ago on Wed 18 Feb 2026 04:34:59 AM EST.
Package dhcp-server-12:4.4.2-20.b1.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[svr1@localhost ~]$
To direct input to this VM, click inside or press Ctrl+G.
26°C
Mostly cloudy
Search
8:53 PM
2/18/2026



3. Configure and Start the DHCP Service



4. Verify DHCP Client IP Assignment (Fedora)

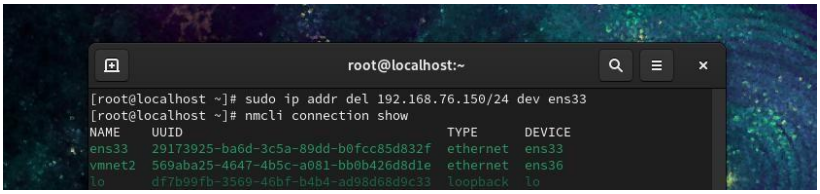
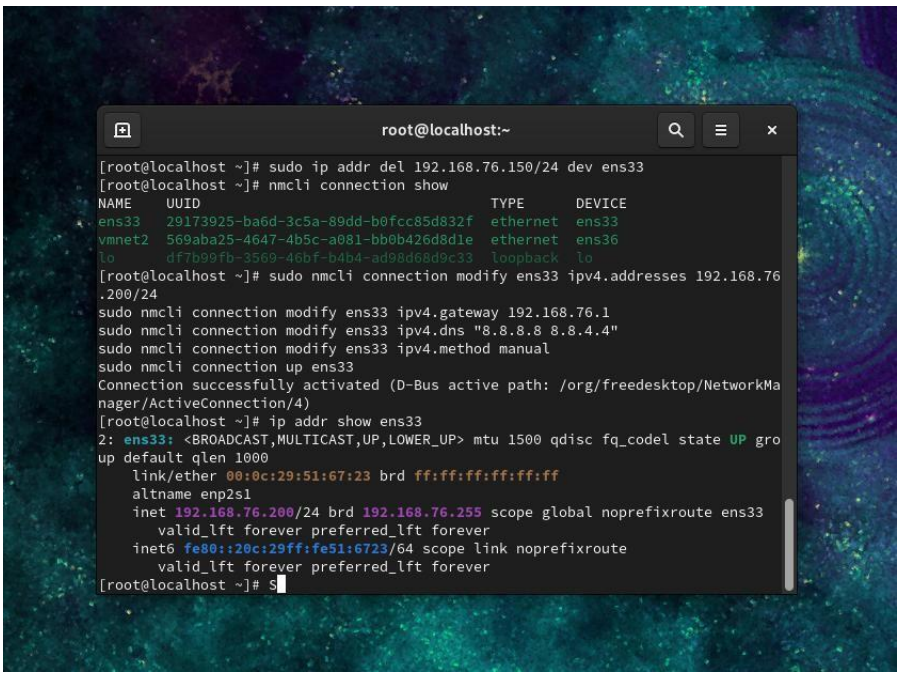
- Verify IP via nmcli

```
AL.DEVICE:                          ens160
AL.TYPE:                            ethernet
AL.HWADDR:                          00:0C:29:D7:E2:CF
AL.MTU:                             1500
AL.STATE:                           100 (connected)
AL.CONNECTION:                      Wired connection 1
AL.CON-PATH:                        /org/freedesktop/NetworkManager/ActiveC
-PROPERTIES.CARRIER:                on
DDRESS[1]:                          10.0.1.20/24
ATEWAY:                             10.0.1.1
OUTE[1]:                            dst = 10.0.1.0/24, nh = 0.0.0.0, mt = 1
OUTE[2]:                            dst = 0.0.0.0/0, nh = 10.0.1.1, mt = 20
NS[1]:                              8.8.8.8
OMAIN[1]:                           dsnm.sub
DDRESS[1]:                          fe80::8528:a43e:33a7:b720/64
ATEWAY:                             --
OUTE[1]:                            dst = fe80::/64, nh = ::, mt = 1024

AL.DEVICE:                          lo
AL.TYPE:                            loopback
AL.HWADDR:                          00:00:00:00:00:00
AL.MTU:                             65536
AL.STATE:                           100 (connected (externally))
: 1-23
```

- Ping command

```
[root@localhost ~]# ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=55.0 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=51.7 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=47.0 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=47.8 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 46.951/50.380/55.019/3.218 ms
[root@localhost ~]# S
```

Changing IP Configurations in Linux – Self-study